

Stage-3 Raw CDF Refinement Distillation, Ablation, and Final Metrics

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2026 年 5 月 22 日

Claim: Stage-3 refinement turns the Stage-2 heteroscedastic teacher into a raw-CDF-aware predictor by mixing direct raw supervision with teacher distillation under time-dependent coverage regimes.

Current status after the $\Delta P^{0.5}$ feature-engineering update:

- the teacher/student feature space uses $A_{\text{scale}} = \Delta P^{0.5} \rho_a^{-0.25} d_n^{0.5}$;
- production Stage-3 is evaluated over five seeds: 7/17/42/99/2024;
- the dominant winning ablation is `anchor_off`;
- refreshed full-clean evaluation has 692 942 points and 24 316 trajectories;
- empirical baselines and the Stage 3 SVGP are included for comparison.

Why Stage-3 Exists

Stage-2 teacher

Stage 2 learns a heteroscedastic Gaussian model from fitted representative curves. It is smooth and defined everywhere on 0–5 ms, but it can inherit fitting assumptions from the reduced curve model.

Raw CDF series

Raw CDF observations are closer to measured data, but they are incomplete and time-dependent: coverage thins as trajectories disappear from the field of view or fail quality filters.

Stage-3 refinement

Stage 3 uses raw CDF where it is reliable, teacher distillation where raw data are sparse, and light shape/onset priors to keep trajectories physically plausible.

CDF Coverage Regimes

Time is binned into $\Delta t = 0.1$ ms intervals. For each operating-condition group, smoothed CDF sample counts define three supervision regimes:

Regime	Trigger	Supervision role
raw_reliable	coverage $\geq 70\%$ of peak	raw CDF is trusted directly
raw_uncertain	below 70%, above 20%	teacher dominates transition
teacher_only	below 20% or < 4 samples	teacher is the only usable signal

Transitions require consecutive bins, which prevents isolated coverage dips from changing the regime.

Default Supervision Weights

Regime	Raw NLL weight	KD KL weight
<code>raw_reliable</code>	1.00	0.25
<code>raw_uncertain</code>	0.00	1.00
<code>teacher_only</code>	0.00	0.75

- Raw NLL is used only where raw CDF is reliable and finite.
- KD KL prevents the student from drifting where raw observations are sparse.
- A global inverse-frequency time-bin weight is clipped to 0.5–2.0.
- Teacher confidence further down-weights high- σ teacher regions.

Source: `MLP/MLP_training/train_stage3_distillation_plus_raw_series.py`

$\Delta P^{0.5}$ A-Scaled Physical Space

Current engineered scale:

$$A_{\text{scale}} = \Delta P^{0.5} \rho_a^{-0.25} d_n^{0.5}.$$

Network output is still scaled:

$$\hat{\mu}(t) = \mu(t)/A, \quad \hat{\sigma}(t) = \sigma(t)/A.$$

Stage-3 losses are computed in physical millimetres:

$$\mu_{\text{phys}} = A\hat{\mu}, \quad \sigma_{\text{phys}} = A\hat{\sigma}.$$

This lets raw CDF targets enter directly in mm while keeping the sparse pressure/diameter axes collapsed through the data-supported A -scale.

Source: MLP/MLP_training/engineered_feature_common.py

Composite Loss

$$\mathcal{L} = \mathcal{L}_{\text{raw}} + \mathcal{L}_{\text{KD}} + \lambda_{\text{onset}} \mathcal{L}_{\text{onset}} + \lambda_{\text{anchor}} \mathcal{L}_{\text{anchor}} + \lambda_{d_1} P_{d_1} + \lambda_{d_2} P_{d_2}.$$

Term	Role
Raw NLL	Gaussian NLL against raw CDF in physical space
KD KL	Gaussian KL between student and Stage-2 teacher in physical space
Onset BCE	soft ramp target in the first 0.2 ms
Anchor	optional early-time penalty on $\mu(t \approx 0)$
P_{d_1}	monotonicity penalty against negative increments
P_{d_2}	concavity penalty after $t \geq 0.5$ ms

Loss Terms: Probabilistic Meaning

Raw Gaussian NLL

Where raw CDF samples are reliable, the student is scored directly in physical mm:

$$\mathcal{L}_{\text{raw},i} = \frac{1}{2} \left[\frac{(y_i - \mu_i)^2}{\sigma_i^2} + \log \sigma_i^2 \right] + \text{const.}$$

This rewards both accurate means and correctly sized variances; shrinking σ_i is penalized if residuals remain large.

Teacher-student Gaussian KL

When raw coverage is weak, the student is kept close to the Stage-2 teacher:

$$D_{\text{KL}}(T\|S) = \log \frac{\sigma_S}{\sigma_T} + \frac{\sigma_T^2 + (\mu_T - \mu_S)^2}{2\sigma_S^2} - \frac{1}{2}.$$

The direction $T\|S$ treats the teacher distribution as the reference distribution that the student should cover in sparse regions.

Key Ablation Finding: Anchor Off

Old design assumption

The early anchor was originally included to keep $\mu(t \approx 0)$ close to zero and prevent unphysical onset behavior.

What the ablation found

The best current Stage-3 setting is usually `anchor_off`, i.e. $\lambda_{\text{anchor}} = 0$. In the five-seed $\Delta P^{0.5}$ mode-C pipeline, seeds 42, 17, 99, and 7 selected `anchor_off`; seed 2024 selected `raw_reliable_no_kd`.

Interpretation

The learned onset head, raw/KD regime weights, and shape priors are enough to regularize early time. A hard $\mu(0)$ anchor can overconstrain the raw-CDF refinement.

Metric Contract for Stage-3 Results

Accuracy metrics

RMSE emphasizes large misses; MAE reports the typical absolute millimetre error; P95 absolute error reports the tail. Bias keeps the sign, so it exposes systematic over- or under-prediction that RMSE/MAE hide.

Uncertainty metrics

Coverage is

$$\Pr(|y - \mu| \leq k\sigma), \quad k \in \{1, 2\}.$$

For a calibrated Gaussian, the nominal targets are 0.683 and 0.954. Higher values mean conservative intervals; lower values mean under-coverage.

Protocol separation

The five-seed in-pipeline result measures training stability on each seed's post-evaluation subset. The external clean diagnostic is the headline comparison because every model is evaluated on the same 692 942 full-clean points or the same 542 565 conservatively uncensored CDF points.

Production Five-Seed Repeatability

This is the refreshed production `anchor_off` Stage-3 model evaluated over seeds 7/17/42/99/2024.

Metric	Full clean	CDF uncensored
RMSE mm	4.735 ± 0.037	4.265 ± 0.040
MAE mm	3.293 ± 0.039	3.032 ± 0.048
Bias mm	$+0.626 \pm 0.112$	$+0.562 \pm 0.119$
P95 abs. err. mm	9.167 ± 0.066	8.307 ± 0.076
1σ coverage	0.867 ± 0.004	0.887 ± 0.004
2σ coverage	0.985 ± 0.001	0.989 ± 0.001

The per-seed spread is small; the conclusion is not seed luck. The CDF-uncensored table is the cleaner physical metric, while full-clean preserves continuity with earlier thesis material.

Source: `Thesis/generated/baseline_comparison_20260521/fixed_table_group_summary.csv`

External Clean Diagnostic

The refreshed production Stage-3 models were evaluated on the full clean diagnostic protocol: 692 942 points and 24 316 trajectories.

Model	RMSE	MAE	P95	1σ cov.
Hiroyasu–Arai calibrated	11.268	8.658	22.891	0.615
Naber–Siebers delay	10.545	8.177	21.253	0.637
Production MLP anchor-off, 5-seed mean	4.735	3.293	9.167	0.867
μ -anchor raw-uncertain MLP, 5-seed mean	5.200	3.654	10.525	0.834
Sparse heteroscedastic GP, Stage 3	4.628	3.173	9.242	0.703

The production MLP crushes the physical baselines and is close to SVGP, but the SVGP remains the point-accuracy winner. The MLP contributes conservative coverage and a deployable neural surrogate.

Source: Thesis/generated/baseline_comparison_20260521/headline_full_clean.csv

Clean vs Hard-Subset Calibration

External clean diagnostic

The production Stage-3 MLP over-covers on the clean full diagnostic: $1\sigma/2\sigma$ coverage is 0.867/0.985, above nominal 0.683/0.954.

In-pipeline post-evaluation subset

The same pipeline has $1\sigma/2\sigma$ coverage of 0.654/0.966, much closer to nominal and slightly under at 1σ .

Paper interpretation

The learned σ is not a single global error bar. It behaves differently on clean/easy trajectories and harder representative rows, which is evidence for difficulty-aware uncertainty rather than pure input-independent noise.

Raw Supervision Source Ablation (single run)

Does replacing `series_wide_clean` with the per-condition right-censored `series_wide_truncated` improve generalisation?

Configuration	RMSE	MAE	Bias	P95
Production (<code>series_wide_clean</code>)	4.746	3.290	—	—
Truncated source (<code>series_wide_truncated</code>)	4.743	3.305	+0.751	9.228
Δ (truncated – production)	–0.003	+0.015	—	—

Conclusion. The Δ RMSE is negligible (≈ 0.003 mm). The regime weight mechanism already zeros the raw loss for low-confidence bins; explicitly truncating those frames in the source table is therefore redundant. This serves as a positive control confirming that the regime labelling correctly isolates the unreliable frames.

Source: `MLP/MLP_training/stage3_truncated_supervision_config.json`

Regime Weight Filter Ablation (single run)

What is the cost of *not* zeroing `raw_weight` for the `raw_uncertain` and `teacher_only` regimes?

Configuration	RMSE	MAE	Bias	P95
Production (filter on : weights = 0)	4.746	3.290	—	—
No regime filter (weights = 1.0)	4.811	3.350	+0.721	9.254
Δ (no-filter – production)	+0.065	+0.060	—	+0.026

Conclusion. Admitting right-censored late-time frames as raw supervision degrades RMSE by +0.065 mm ($\approx 1.4\%$). The KD term partially compensates (high KD weights in uncertain bins suppress the noise), limiting the damage but not eliminating it. Zeroing the raw weight in low-confidence bins is therefore a justified and quantifiably beneficial design choice.

Source: `MLP/MLP_training/stage3_no_regime_raw_filter_config.json`

What Should Go Into the Thesis

Method sentence

Stage 3 refines the Stage-2 heteroscedastic teacher against raw CDF measurements using regime-dependent raw NLL and teacher KL losses, with regimes determined by per-condition CDF coverage over time.

Ablation sentence

The best $\Delta P^{0.5}$ Stage-3 configuration removes the explicit early-time anchor in four of five seeds, indicating that the onset head and raw/KD regime weighting provide sufficient early-time regularization. Two additional single-run ablations confirm that the regime filter is effective: replacing the clean source with the right-censored truncated source yields negligible change ($\Delta \text{RMSE} \approx -0.003$ mm), while disabling the regime raw-weight filter degrades RMSE by $+0.065$ mm.

Result sentence

On the refreshed full-clean benchmark, the resulting five-seed production MLP reaches RMSE 4.735 ± 0.037 mm and outperforms Hiroyasu–Arai and Naber–Siebers by more than 55% in RMSE. It is slightly behind the Stage 3 SVGP in point RMSE, but gives more conservative coverage.

Data Sources for Thesis Extraction

Item	Path
Stage-3 training script	MLP/MLP_training/train_stage3_distillation_plus_raw_series.py
Production MLP runs	MLP/runs_mlp/distill_cdf_onset_v2_ablate_anchor_off_20260521_122435-..._124004
Fixed-table comparison	MLP/baseline/comparison_reports/stage3_fixed_table_eval_with_ha_ns_20260521/
Full-clean comparison table	Thesis/generated/baseline_comparison_20260521/headline_full_clean.csv
SVGP Stage 3 output	MLP/runs_mlp/gp_baseline_stage3_20260521_112229/